

*Direction Régionale : Casablanca – Settat
Nouaceur*

Cité des métiers et des compétences

**Examen de fin de module (EFM) : GESTION DES PROJETS (M109)
Année 2025/2026**

Filière : Intelligence Artificielle

Niveau : TS

Groupe : IA104

Nom et Prénom :

Durée : 2H

Variante : A

Barème : .../20

Theoretical Part :

Multiple Choice Questions (10 points) — Select the one correct answer for each question.

1. A development team wants to download a complete copy of a project hosted on GitHub in order to start working locally. Which command should they use? (1.25 pts)
 - (a) git push
 - (b) git clone
 - (c) git commit
 - (d) git merge
2. A company is developing an AI platform consisting of several modules (Computer Vision, NLP, and API). In SAFe, this overall initiative will be divided into several functionalities. What is this initiative called? (1.25 pts)
 - (a) User Story
 - (b) Sprint
 - (c) Epic
 - (d) Feature
3. During Sprint planning, the team selects the User Stories it commits to completing over the next two weeks. This meeting is called : (1.25 pts)
 - (a) Daily Scrum
 - (b) Sprint Review
 - (c) Sprint Planning
 - (d) Sprint Retrospective
4. A developer wants to describe a feature from the user's perspective so that the team clearly understands the business requirement. Which artifact should be created? (1.25 pts)
 - (a) Test Plan
 - (b) Gantt Chart
 - (c) User Story
 - (d) UML Diagram
5. Before pushing changes to GitHub, a developer wants to retrieve the latest updates made by teammates to avoid merge conflicts. Which command should be used? (1.25 pts)
 - (a) git clone
 - (b) git commit
 - (c) git pull
 - (d) git init
6. In SAFe, a functionality that is small enough to be developed during a Program Increment is called a : (1.25 pts)
 - (a) Epic
 - (b) Feature
 - (c) Sprint Goal
 - (d) User Story
7. During the Daily Scrum, two developers begin discussing software architecture for several minutes. According to Scrum, what is the best practice? (1.25 pts)
 - (a) Continue the discussion until a solution is found.
 - (b) Postpone the discussion until after the Daily Scrum with the relevant team members.
 - (c) Cancel the meeting.
 - (d) Ask the Product Owner to make the decision immediately.
8. Which principle is shared by both Scrum and SAFe? (1.25 pts)
 - (a) They use iterative and incremental development cycles.
 - (b) They require planning only at the beginning of the project.
 - (c) They prohibit synchronization meetings.
 - (d) They completely replace the Product Owner role.

Practical Part : Planning an Artificial Intelligence Platform for Predictive Maintenance. (30 points)

An industrial company plans to develop an intelligent platform capable of predicting machine failures using Machine Learning and Deep Learning techniques.

The project manager decides to use the **PERT** method to plan project activities, analyze task dependencies, identify the critical path, and estimate the project's minimum completion time.

After several meetings involving the Data Engineering, AI, and Software Development teams, the following tasks have been identified.

Code	Task Description	Duration	Predecessor(s)
A	Requirements analysis and specification writing	3	–
B	IoT sensor data collection	6	A
C	Platform architecture design	5	A
D	Data preprocessing and cleaning	5	B
E	Feature selection	4	D
F	AI model development	8	D
G	Integration API development	6	C
H	Model validation and optimization	5	E,F
I	Dashboard development	4	G
J	Integration of the API with the AI model	4	G,H
K	Functional and technical testing	5	I,J
L	Platform deployment	3	K
M	Documentation and user training	4	L

Based on the project above, **circle the one correct answer**.

- Which task has two immediate successors? (1 pt)
 - C
 - H
 - D
 - G
- Which task immediately precedes the deployment phase? (2 pts)
 - I
 - J
 - K
 - M
- What is the earliest start time of task D? (1 pt)
 - 3
 - 11
 - 6
 - 9
- Which task has two predecessors? (1 pt)
 - E
 - H
 - F
 - G
- Which task can be executed in parallel with task D? (1 pt)
 - E
 - G
- What is the earliest finish time of task F? (1 pt)
 - 14
 - 22
 - 24
 - 19
- The critical path is : (1 pt)
 - A-B-D-E-H-J-K-L-M
 - A-C-G-J-K-L-M
 - A-B-D-F-H-J-K-L-M
 - A-C-G-I-K-L-M
- How many tasks belong to the critical path? (1 pt)
 - 7
 - 8
 - 9
 - 10
- Which task has the longest duration? (2 pts)
 - G
 - H
 - F
 - B
- A task with a total float equal to zero is : (1 pt)
 - An optional task
 - An independent task
 - A parallel task
 - A critical task

11. Which task has a positive float ? (2 pts)
- (a) F
 - (b) H
 - (c) G
 - (d) L
12. Which task is the convergence point of the "Feature Selection" and "AI Model Development" branches ? (2 pts)
- (a) H
 - (b) I
 - (c) J
 - (d) K
13. If the duration of task F increases by 3 days, the total project duration becomes : (2 pts)
- (a) 46 days
 - (b) 47 days
 - (c) 44 days
 - (d) 48 days
14. Which task depends only on the software architecture ? (2 pts)
- (a) G
 - (b) H
 - (c) J
 - (d) I
15. What is the earliest possible start time of task G ? (1 pt)
- (a) Day 3
 - (b) Day 5
 - (c) Day 8
 - (d) Day 10
16. Which is the final task of the project ? (2 pts)
- (a) K
 - (b) L
 - (c) M
 - (d) J
17. The primary purpose of the PERT method is to : (2 pts)
- (a) Design databases without project planning.
 - (b) Identify critical tasks and plan the project.
 - (c) Develop a REST API.
 - (d) Perform unit testing and project planning.
18. What is the total duration of the project ? (1 pt)
- (a) 38 days
 - (b) 41 days
 - (c) 43 days
 - (d) 45 days
19. If task G is delayed by two days, what is the impact on the project ? (2 pts)
- (a) No impact.
 - (b) The project will be delayed by two days.
 - (c) Only task I will be delayed.
 - (d) It cannot be determined without calculating the floats.
20. Which statement is correct ? (2 pts)
- (a) All tasks have zero float.
 - (b) All critical tasks have zero float.
 - (c) All tasks with two predecessors are critical.
 - (d) Parallel tasks are always critical.